

EXPERT SYSTEM USING PROLOG

1. Introduction

An expert system is a computer program that uses knowledge and logical rules to solve problems or make decisions like a human expert. It is one of the important areas in Artificial Intelligence because it helps users to receive advice or solutions based on the information they provide.

This practical work focuses on the development of a simple disease diagnosis expert system using SWI-Prolog. The system asks the user questions about symptoms and uses rules to determine a possible disease. The diseases considered in the system include malaria, typhoid, common cold, flu, and food poisoning.

2. Objective of the Practical

The main objective of this practical is to create and test a simple expert system using Prolog.

The specific objectives are:

1. To install and set up SWI-Prolog.
2. To create a Prolog file with the extension `.pl`.
3. To write facts, rules, and questions in Prolog.
4. To load and run the Prolog file.
5. To test the expert system by answering questions.
6. To capture screenshots of the system during execution.

3. Tools Used

The following tools were used for the practical:

- SWI-Prolog
- VS Code / Notepad
- Screenshot tool
- Microsoft Word

4. Installation and Environment Setup

SWI-Prolog was installed on the computer using the official SWI-Prolog installer. The installation was done using the default settings. After installation, SWI-Prolog was opened and tested to ensure that it was working correctly.

A simple test command was typed in SWI-Prolog to confirm that the environment was ready for use.

Insert Screenshot 1: SWI-Prolog opened successfully

5. Creation of the Prolog File

After setting up SWI-Prolog, a new Prolog file was created and saved with the name:

```
disease_diagnosis.pl
```

The `.pl` extension shows that the file is a Prolog source file. The file contains the facts, rules, and questions used by the expert system.

Insert Screenshot 2: The Prolog code in VS Code or Notepad

6. Description of the Expert System

The expert system developed is a disease diagnosis system. It works by asking the user a series of questions about symptoms. The user answers each question with either `yes` or `no`.

Based on the answers provided, the system applies Prolog rules to identify a possible disease. For example, if the user has fever, headache, chills, sweating, and body pain, the system diagnoses malaria.

The system can diagnose the following conditions:

- Malaria
- Typhoid
- Common cold
- Flu
- Food poisoning
- Unknown disease

If the user's symptoms do not match any of the rules, the system displays `unknown_disease` and advises the user to consult a medical professional.

7. Facts, Rules, and Questions

In the expert system, the user's answers are stored as known facts. These facts are used by the system to decide whether a symptom is present or not.

The rules are written to represent the conditions for diagnosing each disease. For example, malaria is diagnosed when the user has symptoms such as fever, headache, chills, sweating, and body pain.

The system asks questions such as:

- Do you have fever?
- Do you have headache?
- Do you have chills?
- Do you have cough?

- Do you have stomach pain?
- Do you have vomiting?

The answers to these questions help the system to reach a final diagnosis.

8. Loading the Prolog File

The Prolog file was loaded into SWI-Prolog using the consult option or by typing the file name in the Prolog terminal. After the file was loaded successfully, SWI-Prolog displayed `true`, showing that there were no errors in the code.

Insert Screenshot 3: Prolog file loaded successfully

9. Running the Expert System

To run the expert system, the command below was typed in SWI-Prolog:

```
main.
```

After running the command, the system displayed the title of the disease diagnosis system and started asking questions. The user answered the questions by typing `yes.` or `no.`.

For example:

```
Do you have fever? yes.  
Do you have headache? yes.  
Do you have chills? yes.  
Do you have sweating? yes.  
Do you have body_pain? yes.
```

After receiving the answers, the system displayed the final diagnosis.

Insert Screenshot 4: Questions asked by the system

10. Output of the System

After answering the questions, the expert system displayed the final diagnosis and advice. For example, when the symptoms matched malaria, the system produced the following result:

```
Final Diagnosis: malaria
```

The system also displayed advice telling the user to visit a hospital or clinic for a malaria test and treatment.

Insert Screenshot 5: Final diagnosis output

11. Importance of the Expert System

This expert system is important because it shows how Artificial Intelligence can be used to assist in decision-making. Although the system is simple, it demonstrates how Prolog can use logical rules to solve a real-life problem.

The system can help users understand possible causes of their symptoms. However, it should not replace professional medical advice. A qualified health worker should always be consulted for proper diagnosis and treatment.

12. Challenges Encountered

During the practical, one of the main things to note was that Prolog commands must always end with a full stop. Also, when answering questions, the user must type `yes.` or `no.` with a full stop. If the full stop is not added, the command will not run properly.

Another important point was making sure that the file was saved with the correct `.pl` extension and not as a text file.

13. Conclusion

In conclusion, this practical work demonstrated how to set up SWI-Prolog and use it to create a simple expert system. The disease diagnosis system was able to ask questions, receive answers, apply rules, and produce a final diagnosis.

The practical helped to show how facts and rules are used in Prolog to solve problems. It also showed the importance of expert systems in Artificial Intelligence, especially in areas where decision-making is required.